C++

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Different roles

End User : Who is not a programmer. Who executes your programs. Who is non-technical.



Programmer User : Who writes main( ) and does programming for interaction with end user, who uses class library, call functions, create menus, thinks about end user



Developer User : Who designs classes, functionalities with classes and thinks about programmer user. He is senior and experienced.



Procedural Oriented Programming(POP)	Object Oriented Programming(OOP)
Problem is divided in to small logically independent tasks called as procedures or functions.	Problem is divided in to small real time entities called as objects
Importance is given to different functions and the sequence in which these functions are called	Importance is given to data and not to functions, actually functions are called on real time entities i.e. objects
We use only global and local scope. Does not have any access specifiers as such.	There are 3 access specifiers in C++ public, private and protected
Data is not hidden.	Data may be kept hidden by making it private
Examples : C, COBOL, PASCAL, FORTRAN etc.	Examples: C++, Java, C#, SmallTalk etc.

class

1. Class is classification of an entity
2. It is user-defined data type
3. It is collection of data members and member functions
4. It is blue-print for an object.
5. It is template for an object.
6. We declare class once in our program and create many objects of that class
7. Class is best example of encapsulation
8. At the time defining a class we do data level abstraction.
9. It is a logical entity. (concept)

object

1. It is a variable of type class
2. It is representation of an entity which has state, behavior and identity
3. object's life cycle - object is born, it undergoes many changes, it dies.
4. It is physical entity. (memory is allocated for object when created and memory is released when object goes out of scope).
5. We can create multiple objects of a class. Each object will have its own life-cycle.
6. Value of member variable gives us state of object at any given point of time.
7. We call functions of class with object and state of object is available in function.

Abstraction

Abstraction means showing only essential information while hiding the implementation details.

For example, a person driving a car. The driver knows that pressing the accelerator increases the car's speed, and applying brakes stops it. However, the driver doesn't need to understand the complicated inner workings of the car's mechanisms.

Encapsulation

To keep data and functionalities together in one single unit.

Encapsulation is achieved through private and public access specifiers.

Data members (variables) are kept private and member functions are kept public.

my_laptop.on() - When the function / method is called on a particular object, the state of that object is going to change.

The meaning of **Encapsulation**, is to make sure that "sensitive" data is hidden from users.

setter and getter methods

These methods are written for all possible member variables.

setter methods are used to set values of member variables.

getter methods are used to get values of member variables.

Sample code to understand and write class date

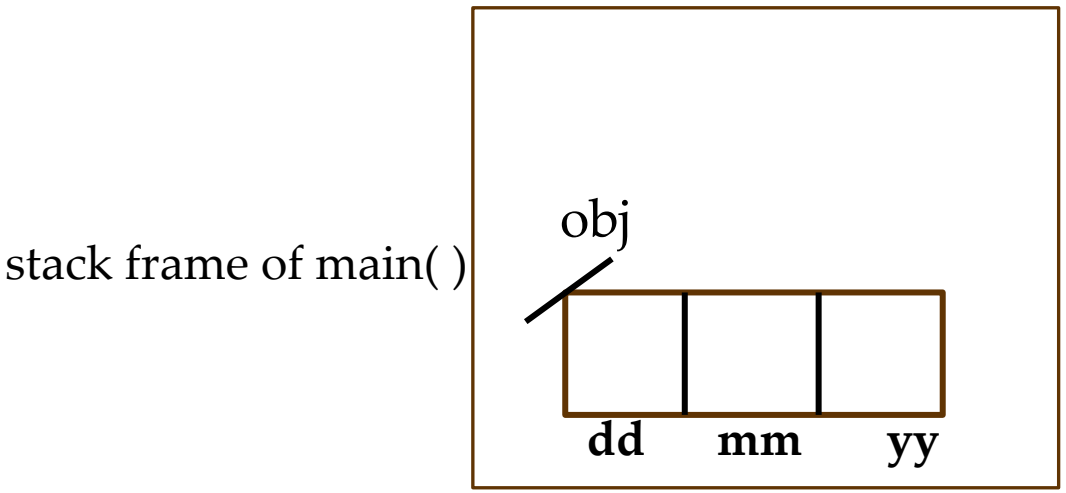
```
#include<iostream>
using namespace std;

class date {
  int dd, mm, yy;

public:
  void set_date(int d, int m, int  y)
   {
      dd = d;
      mm = m;
      yy = y;
   }
  void display( )
   {
      cout << "\n"<< dd << "/" << mm <<" / "<<yy;
   }
};  //end of class
```

```
int main( )
{
    date obj;
    obj.set_date(20,10,2023);
    obj.display( );
    return 0;
}
```

object allocation on stack in RAM



Sample code to understand and write class **date**

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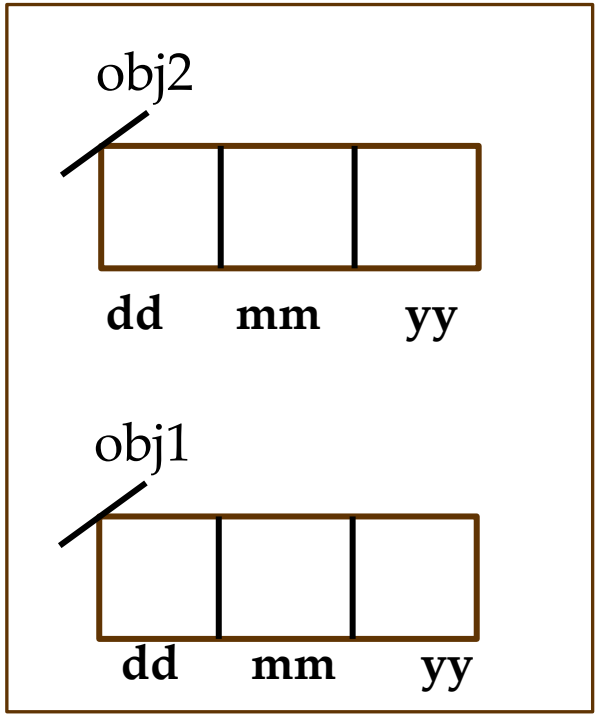
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  void display( )
   {
      cout << "\n"<< dd << "/" << mm <<" / "<<yy;
   }
};  //end of class
```

```
int main( )
{
    date obj1, obj2;
    obj1.set_date(20,10,2023);
    obj1.display( );
    obj2.set_date(23,10,2023);
    obj2.display( );
    return 0;
}
```

object allocation on stack in RAM

stack frame of main( )



Assignment: Do it yourself.

1. Declare and define class time having hh,mm,ss. Write accept_time( ) and display_time( ) in format " hh : mm : ss "
2. Declare and define class point having x, y. Write set_point(int, int) and display_point( ) in format (x , y)